

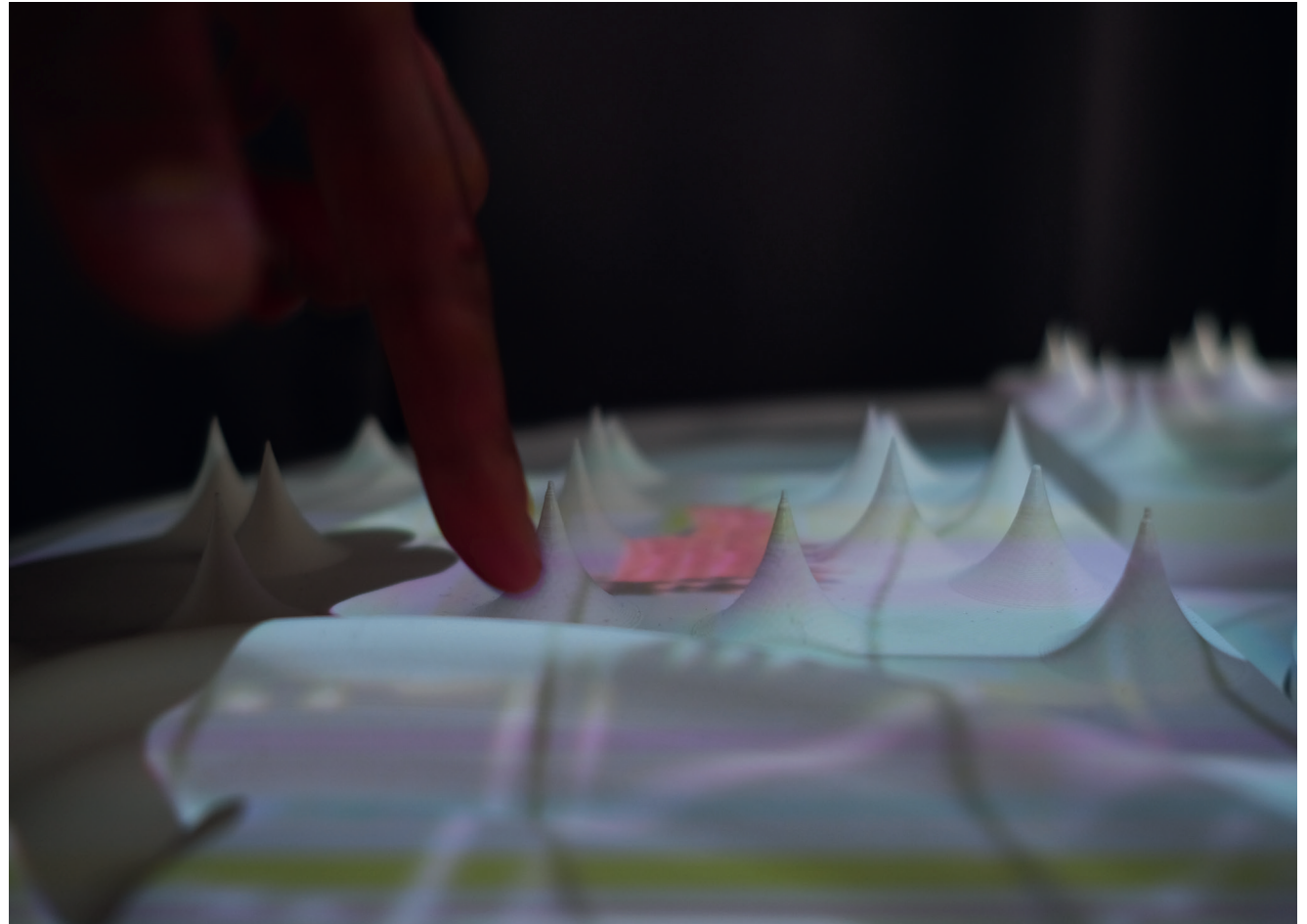
Touching Sound

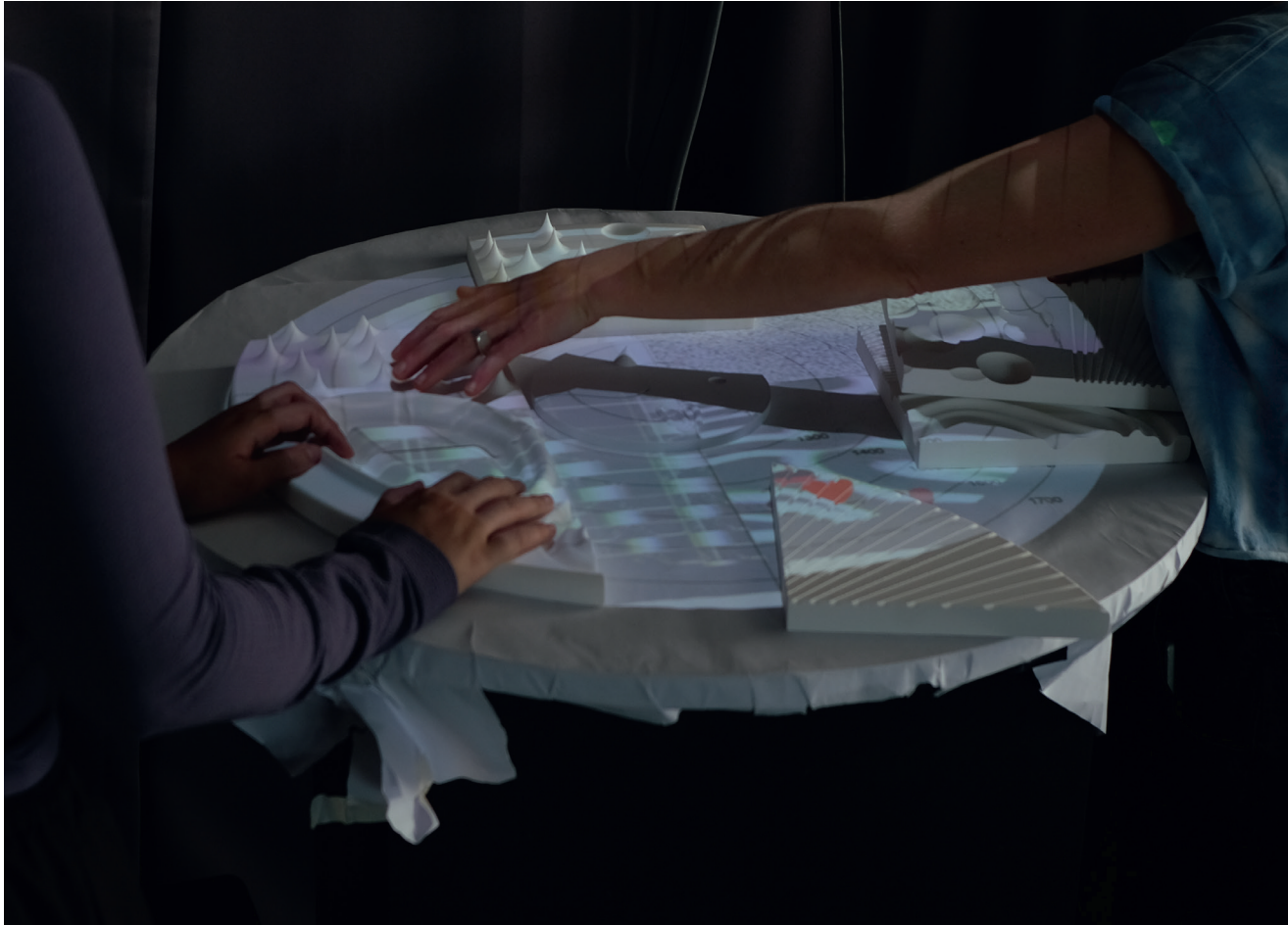
A tactile and sonic journey through suit symbol

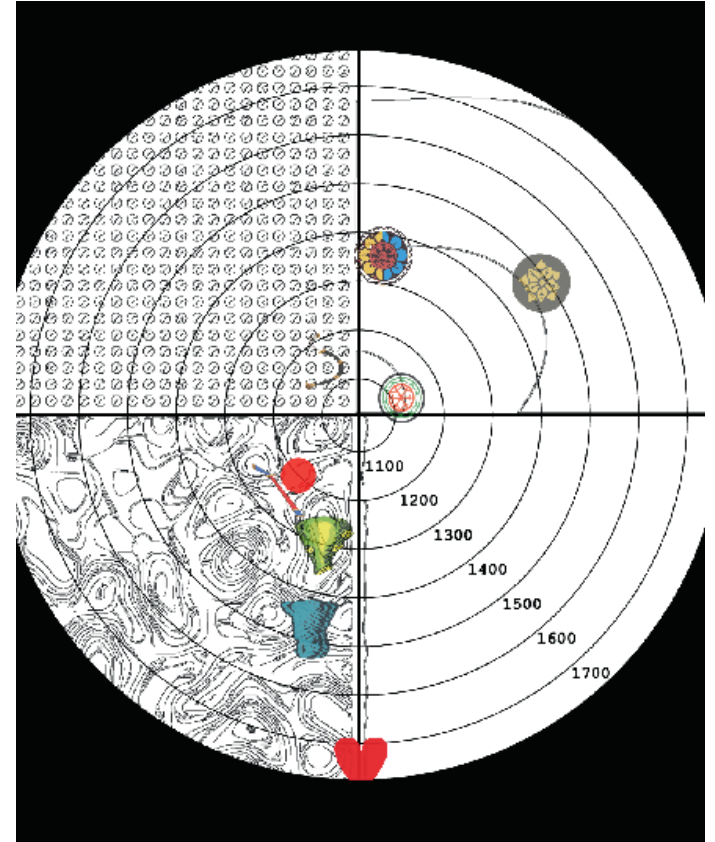
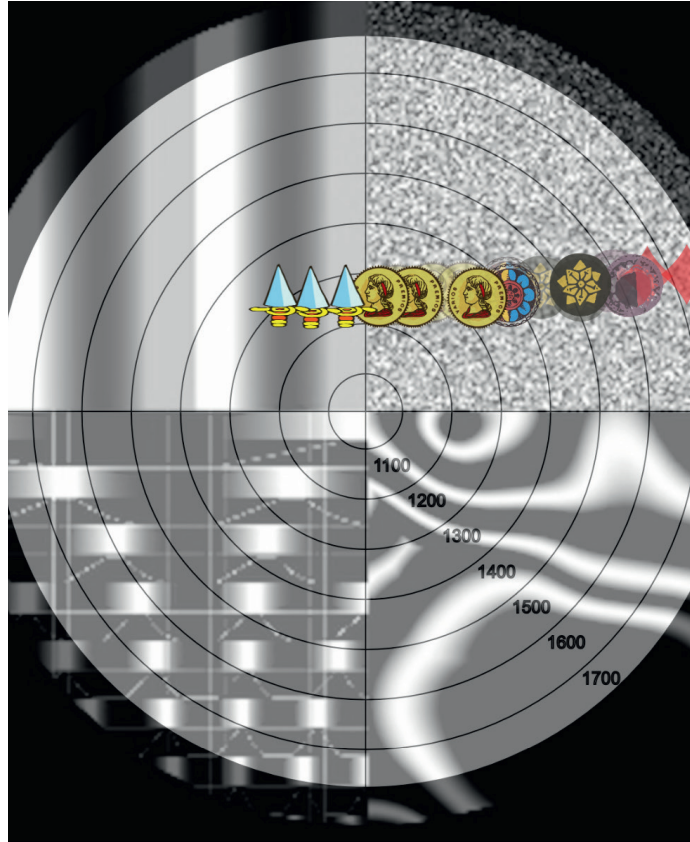
Touching Sound

Touching Sound is an interactive audiovisual installation created in collaboration with Musée de Jeux, exploring the history of card suit symbolism through touch and sound. The four modern French suits (hearts, spades, clubs, and diamonds) trace back to ancient symbols that evolved across cultures and centuries.

Visitors place their hands on a circular 3D-textured table whose surface recalls the shapes of historical suits. As they slide their hands along the reliefs, real-time projections and atmospheric sounds unfold, turning gestures into a sensory dialogue between tactility, imagery, and rhythm. Each movement uncovers the metamorphosis of specific designs, offering a playful, sonic journey through the history of card symbols. Up to four participants can play together, shaping visuals and sound through touch.

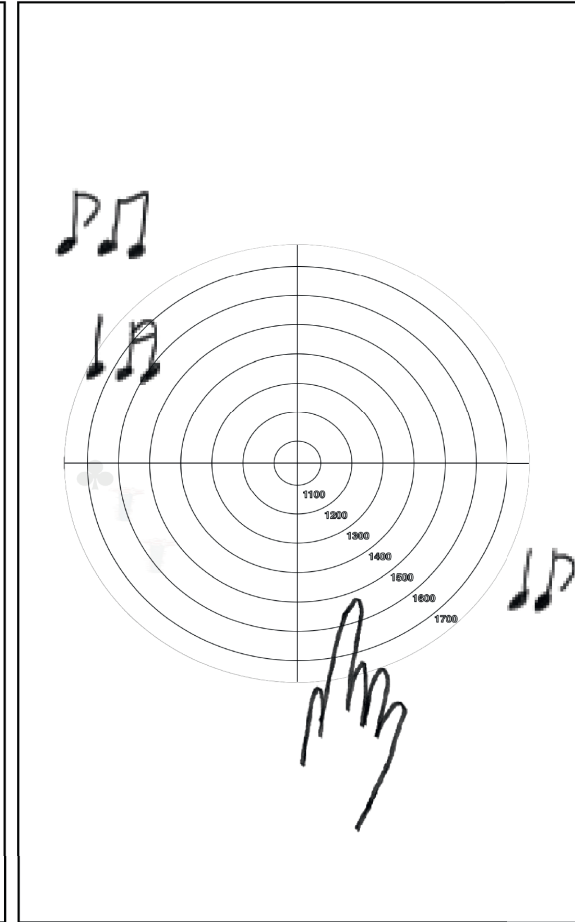
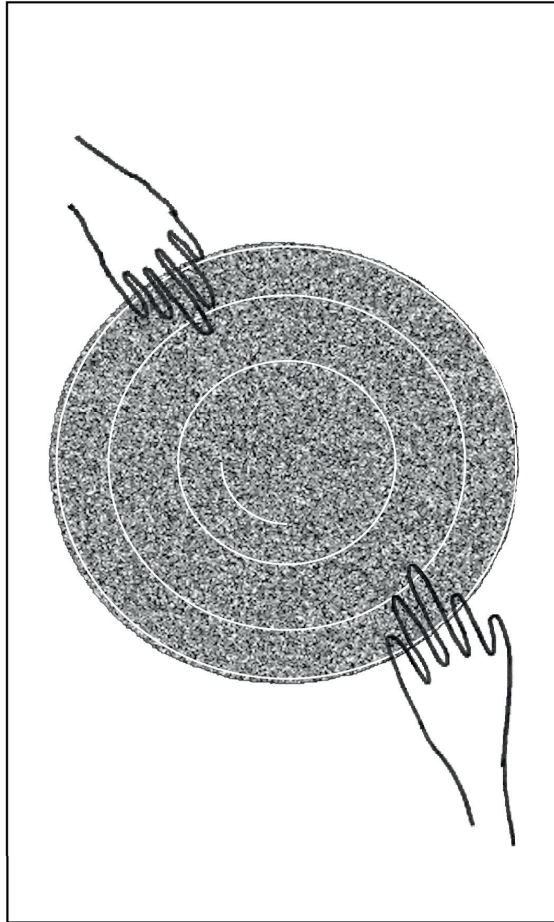
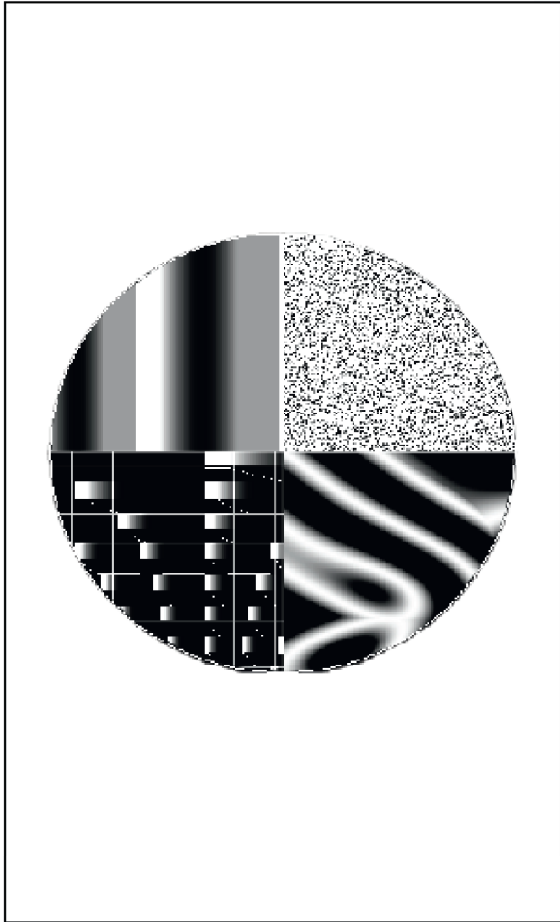






User Journey

1. Visitors encounter a video projected onto a round table divided into four quadrants—each representing a suit—accompanied by an original soundscape inviting them to gather.
2. Concentric lines mark centuries from the 12th to 18th; touching the 3D surface lights up a quadrant and tracks hand movement. Each suit offers a brief sound-reactive interaction tied to its history.
3. After all suits are explored, the table activates for collaborative play, letting up to four users create melodies and visuals together.



Field observations

We observed children and families exploring museums through touch and sound, inspiring our multisensory project. Influenced by Pinaffo & Pluvinaige’s Jelly (2018) and MIT’s The Sound of Touch, we transformed sensory experiences from touch to sound. Visual symbols of historical suits were paired with characteristic textures, tracing their evolution from 12th-century Chinese materials—leather, wood, shell, and metal—to the four French suits. Research on sound-image associations guided the translation of playing card history into sound and visuals.



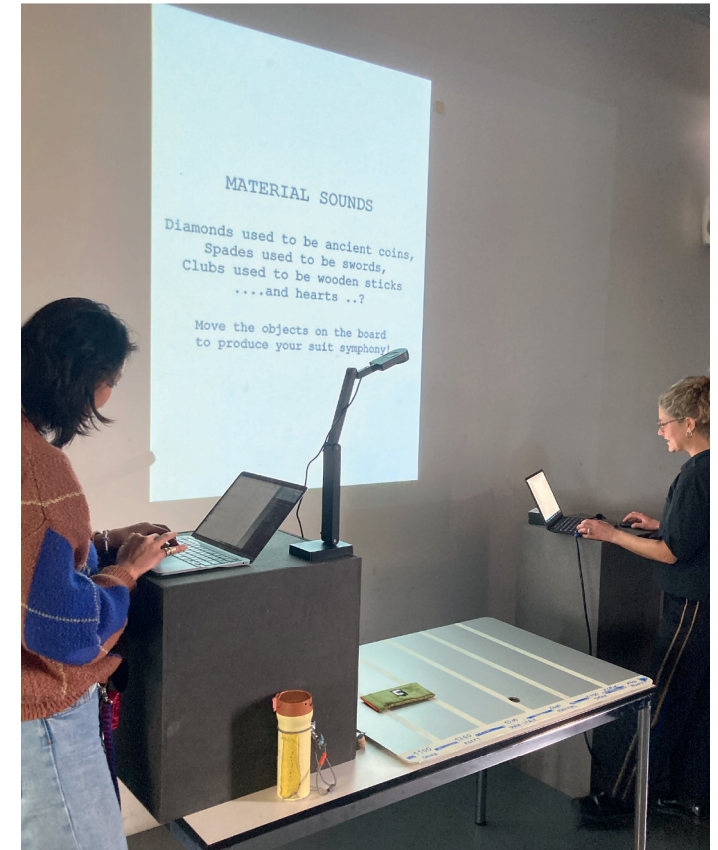
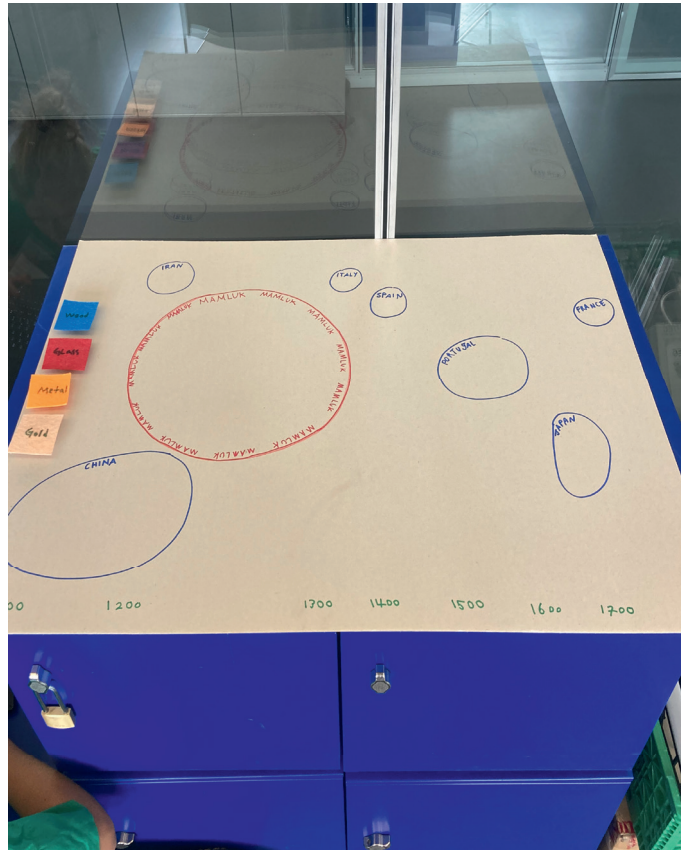
Chinese	Coins 錢	Strings of Cash 串	Myriads of Strings 万	Tens of Myriad 十
Italian	Coins 	Clubs 	Cups 	Sword
Spanish	Coins 	Clubs 	Cups 	Sword
Swiss-German	Bells 	Acorns 	Roses 	Shield
German	Bells 	Acorns 	Hearts 	Leaves
French	Diamonds 	Clubs 	Hearts 	Spade

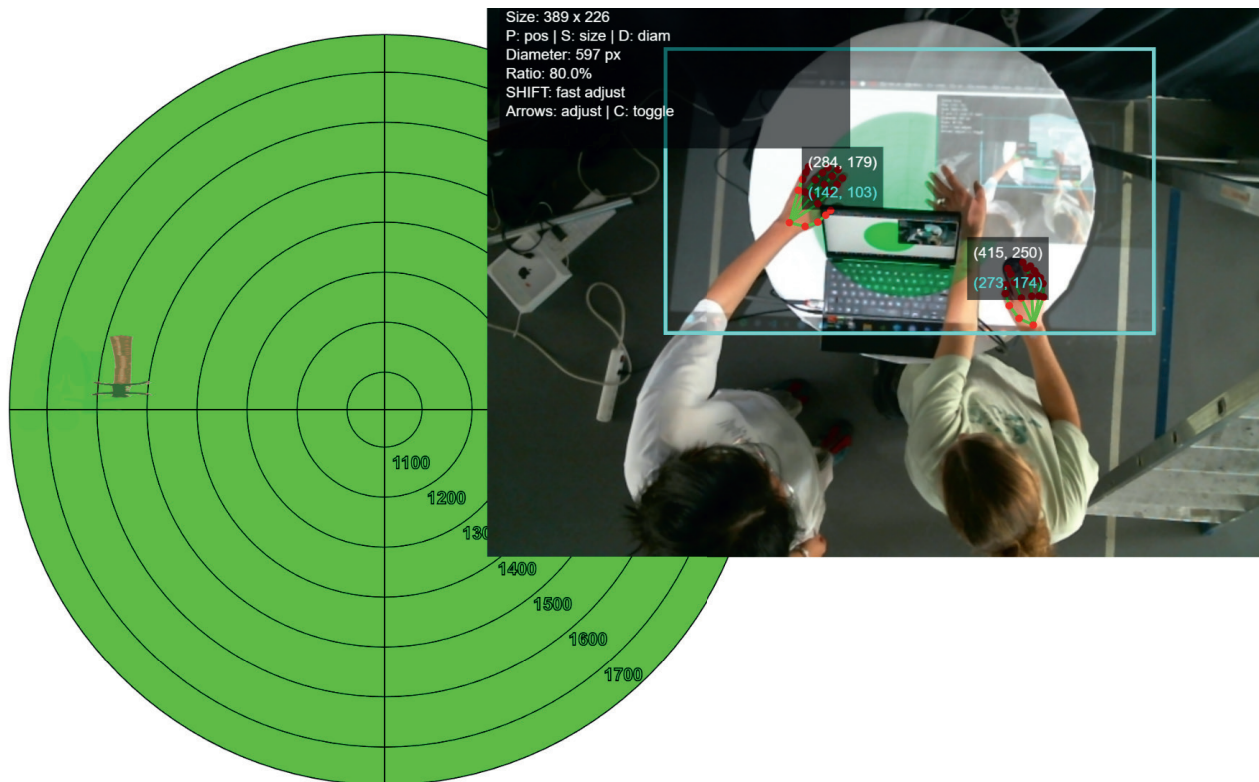
User Tests

Prototype 1 – Rectangular board with ArUco QR codes; sounds changed with X-Y movement via OpenCV. Users tapped instead of sliding, interrupting sound continuity.

Prototype 2 — Magnetic board enabled smoother motion and added ambient sounds, a map timeline, and an info screen—though onboarding was unclear and the screen drew little attention.

Prototype 3 — Circular board introduced multiuser play and real-time finger tracking via p5.js. The 3D-printed PLA surface replaced costly CNC milling, while Gibberish-generated sounds evolved fluidly with hand movement for a more immersive, playful experience.



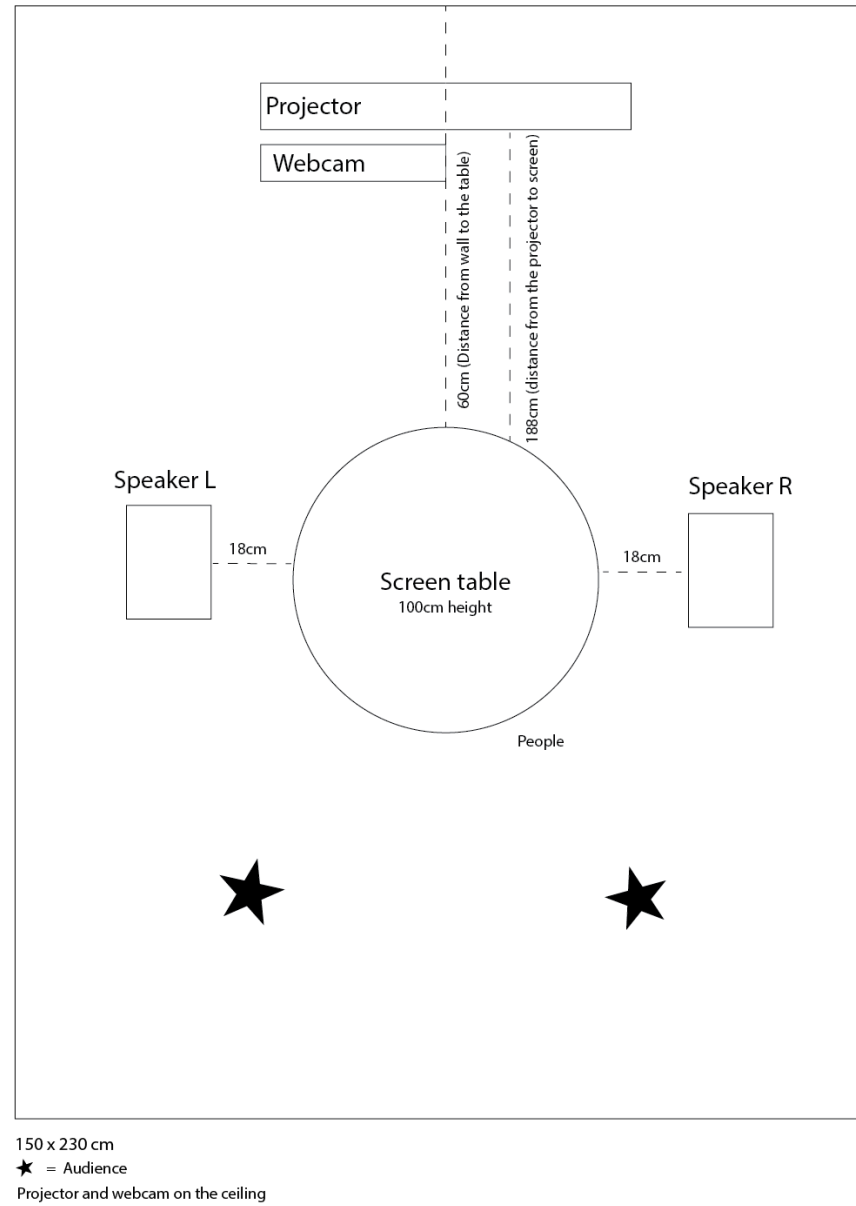


Left: Q4
Right: Q2

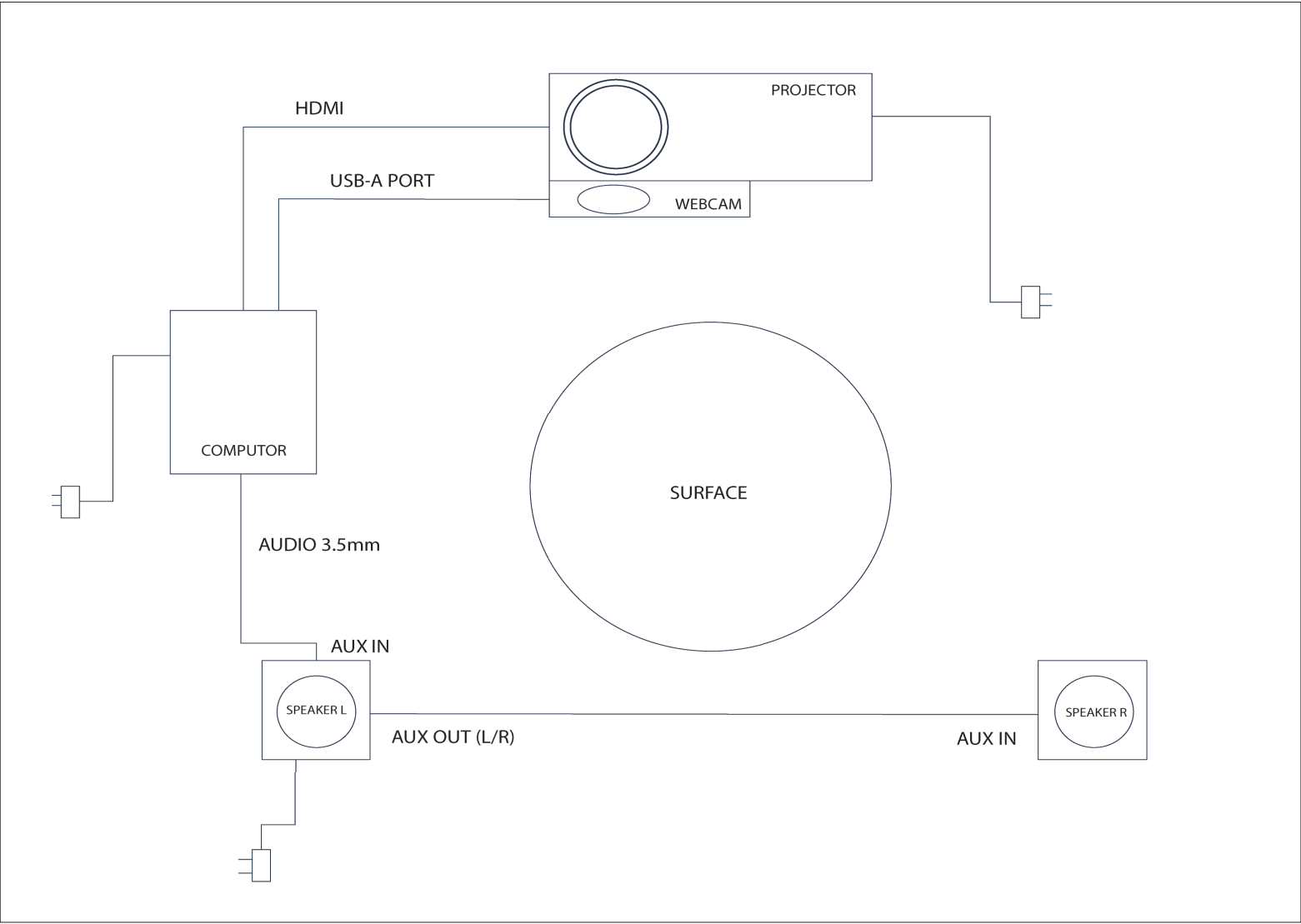




Spatial diagram



Technical diagram



Components List

- Projector
- Webcam
- HDMI cable
- USB-A port
- Computer
- 3.5 mm audio (AUX) cable
- AUX input/output cable
- Pair of speakers